

APPENDIX A
GEOMETRICS BORING LOGS

GROUP SYMBOLS AND NAMES			
Graphic / Symbol	Group Names	Graphic / Symbol	Group Names
	GW Well-graded GRAVEL Well-graded GRAVEL with SAND		CL Lean CLAY Lean CLAY with SAND Lean CLAY with GRAVEL SANDY lean CLAY SANDY lean CLAY with GRAVEL GRAVELLY lean CLAY GRAVELLY lean CLAY with SAND
	GP Poorly graded GRAVEL Poorly graded GRAVEL with SAND		
	GW-GM Well-graded GRAVEL with SILT Well-graded GRAVEL with SILT and SAND		CL-ML SILTY CLAY SILTY CLAY with SAND SILTY CLAY with GRAVEL SANDY SILTY CLAY SANDY SILTY CLAY with GRAVEL GRAVELLY SILTY CLAY GRAVELLY SILTY CLAY with SAND
	GW-GC Well-graded GRAVEL with CLAY (or SILTY CLAY) Well-graded GRAVEL with CLAY and SAND (or SILTY CLAY and SAND)		
	GP-GM Poorly graded GRAVEL with SILT Poorly graded GRAVEL with SILT and SAND		ML SILT SILT with SAND SILT with GRAVEL SANDY SILT SANDY SILT with GRAVEL GRAVELLY SILT GRAVELLY SILT with SAND
	GP-GC Poorly graded GRAVEL with CLAY (or SILTY CLAY) Poorly graded GRAVEL with CLAY and SAND (or SILTY CLAY and SAND)		
	GM SILTY GRAVEL SILTY GRAVEL with SAND		OL ORGANIC lean CLAY ORGANIC lean CLAY with SAND ORGANIC lean CLAY with GRAVEL SANDY ORGANIC lean CLAY SANDY ORGANIC lean CLAY with GRAVEL GRAVELLY ORGANIC lean CLAY GRAVELLY ORGANIC lean CLAY with SAND
	GC CLAYEY GRAVEL CLAYEY GRAVEL with SAND		
	GC-GM SILTY, CLAYEY GRAVEL SILTY, CLAYEY GRAVEL with SAND		OL ORGANIC SILT ORGANIC SILT with SAND ORGANIC SILT with GRAVEL SANDY ORGANIC SILT SANDY ORGANIC SILT with GRAVEL GRAVELLY ORGANIC SILT GRAVELLY ORGANIC SILT with SAND
	SW Well-graded SAND Well-graded SAND with GRAVEL		
	SP Poorly graded SAND Poorly graded SAND with GRAVEL		CH Fat CLAY Fat CLAY with SAND Fat CLAY with GRAVEL SANDY fat CLAY SANDY fat CLAY with GRAVEL GRAVELLY fat CLAY GRAVELLY fat CLAY with SAND
	SW-SM Well-graded SAND with SILT Well-graded SAND with SILT and GRAVEL		
	SW-SC Well-graded SAND with CLAY (or SILTY CLAY) Well-graded SAND with CLAY and GRAVEL (or SILTY CLAY and GRAVEL)		MH Elastic SILT Elastic SILT with SAND Elastic SILT with GRAVEL SANDY elastic SILT SANDY elastic SILT with GRAVEL GRAVELLY elastic SILT GRAVELLY elastic SILT with SAND
	SP-SM Poorly graded SAND with SILT Poorly graded SAND with SILT and GRAVEL		
	SP-SC Poorly graded SAND with CLAY (or SILTY CLAY) Poorly graded SAND with CLAY and GRAVEL (or SILTY CLAY and GRAVEL)		OH ORGANIC fat CLAY ORGANIC fat CLAY with SAND ORGANIC fat CLAY with GRAVEL SANDY ORGANIC fat CLAY SANDY ORGANIC fat CLAY with GRAVEL GRAVELLY ORGANIC fat CLAY GRAVELLY ORGANIC fat CLAY with SAND
	SM SILTY SAND SILTY SAND with GRAVEL		
	SC CLAYEY SAND CLAYEY SAND with GRAVEL		OH ORGANIC elastic SILT ORGANIC elastic SILT with SAND ORGANIC elastic SILT with GRAVEL SANDY elastic ELASTIC SILT SANDY ORGANIC elastic SILT with GRAVEL GRAVELLY ORGANIC elastic SILT GRAVELLY ORGANIC elastic SILT with SAND
	SC-SM SILTY, CLAYEY SAND SILTY, CLAYEY SAND with GRAVEL		
	TS TOPSOIL		OL/OH ORGANIC SOIL ORGANIC SOIL with SAND ORGANIC SOIL with GRAVEL SANDY ORGANIC SOIL SANDY ORGANIC SOIL with GRAVEL GRAVELLY ORGANIC SOIL GRAVELLY ORGANIC SOIL with SAND
	COBBLES COBBLES and BOULDERS BOULDERS		

DRILLING METHOD SYMBOLS



Auger Drilling



Rotary Drilling



Dynamic Cone
or Hand Driven



Diamond Core

DEFINITIONS FOR CHANGE IN MATERIAL

Term	Definition	Symbol
Material Change	Change in material is observed in the sample or core, and the location of change can be accurately measured.	_____
Estimated Material Change	Change in material cannot be accurately located because either the change is gradational or because of limitations in the drilling/sampling methods used.	- - - - -
Soil/Rock Boundary	Material changes from soil characteristics to rock characteristics.	~~~~~

FIELD AND LABORATORY TESTS

C	Consolidation (ASTM D 2435-04)
CL	Collapse Potential (ASTM D 5333-03)
CP	Compaction Curve (CTM 216 - 06)
CR	Corrosion, Sulfates, Chlorides (CTM 643 - 99; CTM 417 - 06; CTM 422 - 06)
CU	Consolidated Undrained Triaxial (ASTM D 4767-02)
DS	Direct Shear (ASTM D 3080-04)
EI	Expansion Index (ASTM D 4829-03)
M	Moisture Content (ASTM D 2216-05)
OC	Organic Content (ASTM D 2974-07)
P	Permeability (CTM 220 - 05)
PA	Particle Size Analysis (ASTM D136)
PI	Liquid Limit, Plastic Limit, Plasticity Index (AASHTO T 89-02, AASHTO T 90-00)
PL	Point Load Index (ASTM D 5731-05)
PM	Pressure Meter
PP	Pocket Penetrometer
R	R-Value (CTM 301 - 00)
SE	Sand Equivalent (CTM 217 - 99)
SG	Specific Gravity (AASHTO T 100-06)
SL	Shrinkage Limit (ASTM D 427-04)
SW	Swell Potential (ASTM D 4546-03)
TV	Pocket Torvane
UC	Unconfined Compression - Soil (ASTM D 2166-06) Unconfined Compression - Rock (ASTM D 2938-95)
UU	Unconsolidated Undrained Triaxial (ASTM D 2850-03)
UW	Unit Weight (ASTM D 4767-04)
VS	Vane Shear (AASHTO T 223-96 [2004])

SAMPLER GRAPHIC SYMBOLS



Standard Penetration Test (SPT)



Standard California Sampler



Modified California Sampler



Shelby Tube



Piston Sampler



NX Rock Core



HQ Rock Core



Grab/Bulk
Sample



Other (see remarks)

WATER LEVEL SYMBOLS



First Water Level Reading (during drilling)



Static Water Level Reading (after drilling, date)

Geometrics Engineering P.S. Inc.

FIGURE NUMBER

PROJECT NAME

Sea-Tac International Airport
SEA S Concourse

PROJECT NUMBER

BORING RECORD LEGEND #1

CONSISTENCY OF COHESIVE SOILS				
Descriptor	Shear Strength (tsf)	Pocket Penetrometer, PP Measurement (tsf)	Torvane, TV. Measurement (tsf)	Vane Shear, VS. Measurement (tsf)
Very Soft	< 0.12	< 0.25	< 0.12	< 0.12
Soft	0.12 - 0.25	0.25 - 0.50	0.12 - 0.25	0.12 - 0.25
Medium Stiff	0.25 - 0.50	0.50 - 1.0	0.25 - 0.50	0.25 - 0.50
Stiff	0.50 - 1.0	1.0 - 2.0	0.50 - 1.0	0.50 - 1.0
Very Stiff	1.0 - 2.0	2.0 - 4.0	1.0 - 2.0	1.0 - 2.0
Hard	> 2.0	> 4.0	> 2.0	> 2.0

APPARENT DENSITY OF COHESIONLESS SOILS	
Descriptor	SPT N_{60} - Value (blows / foot)
Very Loose	0 - 5
Loose	5 - 10
Medium Dense	10 - 30
Dense	30 - 50
Very Dense	> 50

MOISTURE	
Descriptor	Criteria
Dry	No discernable moisture
Moist	Moisture present, but no free water
Wet	Visible free water

PERCENT OR PROPORTION OF SOILS	
Descriptor	Criteria
Trace	Particles are present but estimated to be less than 5%
Few	5 to 10%
Little	15 to 25%
Some	30 to 45%
Mostly	50 to 100%

PARTICLE SIZE		
Descriptor		Size (in)
Boulder		> 12
Cobble		3 - 12
Gravel	Coarse	3/4 - 3
	Fine	1/5 - 3/4
Sand	Coarse	1/16 - 1/5
	Medium	1/64 - 1/16
	Fine	1/300 - 1/64
Silt and Clay		< 1/300

PLASTICITY OF FINE-GRAINED SOILS	
Descriptor	Criteria
Non-plastic	A 1/8-inch thread cannot be rolled at any water content.
Low	The thread can barely be rolled, and the lump cannot be formed when drier than the plastic limit.
Medium	The thread is easy to roll, and not much time is required to reach the plastic limit; it cannot be rerolled after reaching the plastic limit. The lump crumbles when drier than the plastic limit.
High	It takes considerable time rolling and kneading to reach the plastic limit. The thread can be rerolled several times after reaching the plastic limit. The lump can be formed without crumbling when drier than the plastic limit.

CONSISTENCY OF COHESIVE SOILS VS. N_{60}	
Description	SPT N_{60} (blows / foot)
Very Soft	0 - 2
Soft	2 - 4
Medium Stiff	4 - 8
Stiff	8 - 15
Very Stiff	15 - 30
Hard	> 30

CEMENTATION	
Descriptor	Criteria
Weak	Crumbles or breaks with handling or little finger pressure.
Moderate	Crumbles or breaks with considerable finger pressure.
Strong	Will not crumble or break with finger pressure.

Ref: Peck, Hansen, and Thornburn, 1974, "Foundation Engineering", Second Edition

Note: Only to be used (with caution) when pocket penetrometer or other data on undrained shear strength are unavailable.

Geometrics Engineering P.S. Inc.		FIGURE NUMBER
PROJECT NAME Sea-Tac International Airport SEA S Concourse		PROJECT NUMBER
BORING RECORD LEGEND #2		

CLIENT: AECOM

PROJECT NAME: SEA S Concourse

PROJECT NUMBER: _____

PROJECT LOCATION: _____ SEA

DATE STARTED: 05/17/2024 COMPLETED: 05/18/2024

GROUND ELEVATION: 376.94 HOLE SIZE: 10-inch

DRILLING CONTRACTOR: Cascade Drilling

NORTHING: 163353.79 EASTING: 1276927.11

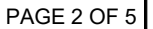
DRILLING METHOD: Mud Rotary GTech Drill GT8 HAMMER WEIGHT: 140-lbs

GROUND WATER LEVELS: _____

SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch AT TIME OF DRILLING: Not Encountered

LOGGED BY: JA CHECKED BY: AB AT END OF DRILLING: Not Encountered

[illegible]



PROJECT NAME:	SEA S Concourse		
PROJECT LOCATION:	SEA		
GROUND ELEVATION:	376.94	HOLE SIZE:	10-inch
NORTHING:	163353.79	EASTING:	1276927.11
GROUND WATER LEVELS:			
AT TIME OF DRILLING:	Not Encountered		
AT END OF DRILLING:	Not Encountered		

[illegible]

CLIENT: AECOM

PROJECT NAME: SEA S Concourse

PROJECT NUMBER: _____

PROJECT LOCATION: SEA

DATE STARTED: 05/17/2024 COMPLETED: 05/18/2024

GROUND ELEVATION: 376.94 HOLE SIZE: 10-inch

DRILLING CONTRACTOR: Cascade Drilling

NORTHING: 163353.79 EASTING: 1276927.11

DRILLING METHOD: HSA HAMMER WEIGHT: 140-lbs

GROUND WATER LEVELS:

SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch AT TIME OF DRILLING: Not Encountered

LOGGED BY: JA CHECKED BY: AB AT END OF DRILLING: Not Encountered

DEPTH (ft)	SAMPLE TYPE NUMBER	BLOW COUNT (N VALUES)	GRAPHIC LOG	MATERIAL DESCRIPTION	LIQUID LIMIT	PLASTICITY INDEX	% GRAVEL	% SAND	% FINES	RECOVERY (IN)	MOISTURE CONTENT (%)	PID (PPM)	REMARKS			
80	S-16	50/4"		Moist, Gray, Hard, Sandy Silt, Trace Fine to Medium Sub-angular Gravel (MH)	25.8				61.8		26.1	0.0				
85	S-17	50/4.5"		Moist, Gray, Very Dense, Silty Fine to Medium Sand, Few Fine to Medium Sub-angular to Sub-rounded Gravel (SM)											0.7	
90	S-18	50/2.5"														0.0
95	S-19	50/2"														
100	S-20	50/3"												0.0		
105	S-21	32-49-50/4"	Moist, Gray, Hard, Clayey SILT with Fine Sand, Few Medium to Coarse Sub Rounded Gravel (ML)			0.0										
110	S-22	50/5"					0.0									
115	S-23	50/2"		Moist, Brown, Very Dense, Well Graded SAND with Silt and Gravel (SW-SM)				23.9	66.3	9.7	2	14.7	0.0			
120																

CLIENT: AECOM

PROJECT NAME: SEA S Concourse

PROJECT NUMBER: _____

PROJECT LOCATION: _____ SEA

DATE STARTED: 05/17/2024 COMPLETED: 05/18/2024

GROUND ELEVATION: 376.94 HOLE SIZE: 10-inch

DRILLING CONTRACTOR: Cascade Drilling

NORTHING: 163353.79 EASTING: 1276927.11

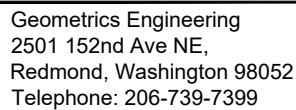
DRILLING METHOD: HSA HAMMER WEIGHT: 140-lbs

GROUND WATER LEVELS: _____

SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch AT TIME OF DRILLING: Not Encountered

LOGGED BY: JA CHECKED BY: AB AT END OF DRILLING: Not Encountered

[illegible]



PAGE 5 OF 5

PROJECT NAME:	SEA S Concourse		
PROJECT LOCATION:	SEA		
GROUND ELEVATION:	376.94	HOLE SIZE:	10-inch
NORTHING:	163353.79	EASTING:	1276927.11
GROUND WATER LEVELS:			
AT TIME OF DRILLING:	Not Encountered		
AT END OF DRILLING:	Not Encountered		

DEPTH (ft)	SAMPLE TYPE NUMBER	BLOW COUNT (N VALUES)	GRAPHIC LOG	MATERIAL DESCRIPTION	LIQUID LIMIT	PLASTICITY INDEX	% GRAVEL	% SAND	% FINES	RECOVERY (IN)	MOISTURE CONTENT (%)	PID (PPM)	REMARKS
160	X S-32	50/2"		Moist, Brown, Very Dense, Silty Fine to Medium SAND (SM)								0.0	
165	Bottom of borehole at 161.5-ft.												
170													
175													
180													
185													
190													
195													
200													



CLIENT: AECOM
PROJECT NUMBER:
DATE STARTED: 05/17/2024 COMPLETED: 05/17/2024
DRILLING CONTRACTOR: Cascade Drilling
DRILLING METHOD: CME 75 HAMMER WEIGHT: 140-lbs
SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch
LOGGED BY: JA CHECKED BY: AB

PROJECT NAME: SEA S Concourse
PROJECT LOCATION: SEA
GROUND ELEVATION: 376.56 HOLE SIZE: 10-inch
NORTHING: 163274.16 EASTING: 1276868.68
GROUND WATER LEVELS:
AT TIME OF DRILLING: Not Encountered
AT END OF DRILLING: Not Encountered

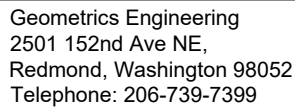
DEPTH (ft)	SAMPLE TYPE NUMBER	BLOW COUNT (N VALUES)	GRAPHIC LOG	MATERIAL DESCRIPTION	LIQUID LIMIT	PLASTICITY INDEX	% GRAVEL	% SAND	% FINES	RECOVERY (IN)	MOISTURE CONTENT (%)	PID (PPM)	REMARKS
				14.25-inch Concrete (PCC)									
				18-inch Aggregate Base Classified as: Moist, Gray, Medium to Coarse Sand, Little Fine to Medium Sub-angular Gravel (SW)								28	
	S-1	25-30-31 (61)		Moist, Gray, Very Dense, Well Graded SAND with Silt (SW-SM)						18		21	
5	S-2	17-33-42 (75)					1.9	88.4	9.7		5.4	273	
	S-3	25-38-47 (85)								18		483	
	S-4	29-50/5" (50/5")		S-4: Trace Coarse Sub-angular Gravel						10		269	
	S-5	31-50/4" (50/4")		S-5: Trace Fine Sub-angular Gravel						10		56	
10	S-6	29-40-50/5"							18.8	14	14.1	203	
	S-7	50-50/3" (50/3")		S-7: Trace Fine Sub-angular Gravel						8		103	
15	S-8	50/5"		S-8: Trace Coarse Sub-angular Gravel						5		49	

Bottom of borehole at 16.5-ft.

CLIENT:	AECOM		
PROJECT NUMBER:			
DATE STARTED:	05/16/2024	COMPLETED:	05/17/2024
DRILLING CONTRACTOR:	Cascade Drilling		
DRILLING METHOD:	HSA	HAMMER WEIGHT:	140-lbs
SAMPLING DEVICE:	Split Spoon 2.5-inch	CASING SIZE:	8-inch
LOGGED BY:	JA	CHECKED BY:	AB

PROJECT NAME:	SEA S Concourse		
PROJECT LOCATION:	SEA		
GROUND ELEVATION:	377.88	HOLE SIZE:	10-inch
NORTHING:	163433.61	EASTING:	1277176.72
GROUND WATER LEVELS:			
AT TIME OF DRILLING:	Not Encountered		
AT END OF DRILLING:	Not Encountered		

[illegible]



PAGE 2 OF 3

LOGGED BY: JA CHECKED BY: AB AT END OF DRILLING: Not Encountered

[illegible]

CLIENT: AECOM

PROJECT NAME: SEA S Concourse

PROJECT NUMBER: _____

PROJECT LOCATION: _____ SEA _____

DATE STARTED: 05/16/2024 COMPLETED: 05/17/2024

GROUND ELEVATION: 377.88 HOLE SIZE: 10-inch

DRILLING CONTRACTOR: Cascade Drilling

NORTHING: 163433.61 EASTING: 1277176.72

DRILLING METHOD: HSA HAMMER WEIGHT: 140-lbs

GROUND WATER LEVELS: _____

SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch AT TIME OF DRILLING: Not Encountered

LOGGED BY: JA CHECKED BY: AB AT END OF DRILLING: Not Encountered

[illegible]



Geometrics Engineering
2501 152nd Ave NE,
Redmond, Washington 98052
Telephone: 206-739-7399

BORING NUMBER S3-5

PAGE 1 OF 1

CLIENT: AECOM
PROJECT NUMBER:
DATE STARTED: 05/16/2024 COMPLETED: 05/16/2024
DRILLING CONTRACTOR: Cascade Drilling
DRILLING METHOD: HSA HAMMER WEIGHT: 140-lbs
SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch
LOGGED BY: AM CHECKED BY: AB

PROJECT NAME: SEA S Concourse
PROJECT LOCATION: SEA
GROUND ELEVATION: 376.98 HOLE SIZE: 10-inch
NORTHING: 163243.20 EASTING: 1277199.20
GROUND WATER LEVELS:
AT TIME OF DRILLING: Not Encountered
AT END OF DRILLING: Not Encountered

DEPTH (ft)	SAMPLE TYPE NUMBER	BLOW COUNT (N VALUES)	GRAPHIC LOG	MATERIAL DESCRIPTION	LIQUID LIMIT	PLASTICITY INDEX	% GRAVEL	% SAND	% FINES	RECOVERY (IN)	MOISTURE CONTENT (%)	PID (PPM)	REMARKS
				14.1-Inch PCC									
				18-inch Aggregate Base Classified as: Moist, Gray, Fine to Medium Sand, Few Fine Sub-angular Gravel (SP)									
	S-1	23-35-45 (80)		Moist, Gray, Very Dense, Silty Fine to Medium SAND, Trace Fine to Medium Sub-angular Gravel (SM)						18		0	
5	S-2	25-50/6" (50/6")								16		0	
	S-3	50/6-50/4 (50/4")					7.1	65.6	27.3	10	8.1	0	
	S-4	26-50-50 (100)		S-4: Gray						18		0	
	S-5	21-30-30 (60)							14.7	18	15.5	0	
10	S-6	28-50/4" (50/4")		Moist, Gray, Very Dense, Silty Fine to Medium Sub-angular Gravel, Few Fine to Medium Sand(GM)			50.9	36.4	12.7	12	7.0	0	
	S-7	45-50/4" (50/4")								10		0	
15	S-8	25-50/3" (50/3")										0	

Bottom of borehole at 16.5-ft.

CLIENT:	AECOM		
PROJECT NUMBER:			
DATE STARTED:	05/15/2024	COMPLETED:	05/16/2024
DRILLING CONTRACTOR:	Cascade Drilling		
DRILLING METHOD:	HSA	HAMMER WEIGHT:	140-lbs
SAMPLING DEVICE:	Split Spoon 2.5-inch	CASING SIZE:	8-inch
LOGGED BY:	JA	CHECKED BY:	AB

PROJECT NAME:	SEA S Concourse		
PROJECT LOCATION:	SEA		
GROUND ELEVATION:	379.27	HOLE SIZE:	10-inch
NORTHING:	163716.01	EASTING:	1277171.43
GROUND WATER LEVELS:			
AT TIME OF DRILLING:	Not Encountered		
AT END OF DRILLING:	Not Encountered		

[illegible]

CLIENT: AECOM

PROJECT NAME: SEA S Concourse

PROJECT NUMBER: _____

PROJECT LOCATION: _____ SEA _____

DATE STARTED: 05/15/2024 COMPLETED: 05/16/2024

GROUND ELEVATION: 379.27 HOLE SIZE: 10-inch

DRILLING CONTRACTOR: Cascade Drilling

NORTHING: 163716.01 EASTING: 1277171.43

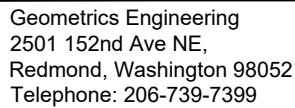
DRILLING METHOD: HSA HAMMER WEIGHT: 140-lbs

GROUND WATER LEVELS: _____

SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch AT TIME OF DRILLING: Not Encountered

LOGGED BY: JA CHECKED BY: AB AT END OF DRILLING: Not Encountered

[illegible]



PAGE 3 OF 5

PROJECT NAME:	SEA S Concourse		
PROJECT LOCATION:	SEA		
GROUND ELEVATION:	379.27	HOLE SIZE:	10-inch
NORTHING:	163716.01	EASTING:	1277171.43
GROUND WATER LEVELS:			
AT TIME OF DRILLING:	Not Encountered		
AT END OF DRILLING:	Not Encountered		

[illegible]

CLIENT: AECOM

PROJECT NAME: SEA S Concourse

PROJECT NUMBER: _____

PROJECT LOCATION: _____ SEA _____

DATE STARTED: 05/15/2024 COMPLETED: 05/16/2024

GROUND ELEVATION: 379.27 HOLE SIZE: 10-inch

DRILLING CONTRACTOR: Cascade Drilling

NORTHING: 163716.01 EASTING: 1277171.43

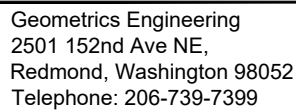
DRILLING METHOD: HSA HAMMER WEIGHT: 140-lbs

GROUND WATER LEVELS: _____

SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch AT TIME OF DRILLING: Not Encountered

LOGGED BY: JA CHECKED BY: AB AT END OF DRILLING: Not Encountered

[illegible]



PAGE 1 OF 1

PROJECT NAME:	SEA S Concourse		
PROJECT LOCATION:	SEA		
GROUND ELEVATION:	379.79	HOLE SIZE:	10-inch
NORTHING:	163959.35	EASTING:	1277075.17
GROUND WATER LEVELS:			
AT TIME OF DRILLING:	Not Encountered		
AT END OF DRILLING:	Not Encountered		

DEPTH (ft)	SAMPLE TYPE NUMBER	BLOW COUNT (N VALUES)	GRAPHIC LOG	MATERIAL DESCRIPTION	LIQUID LIMIT	PLASTICITY INDEX	% GRAVEL	% SAND	% FINES	RECOVERY (IN)	MOISTURE CONTENT (%)	PID (PPM)	REMARKS
0 5 10 15 20				14.05-Inch PCC			45.6	47.4	6.9	16	3.3	0	
		S-1	21-50-50/4" (50/4")	18-inch Aggregate Base Classified as: Moist, Gray, Fine to Medium Sand, Few Fine Sub-angular Gravel (SP)									
		S-2	30-50/5" (50/5")	Moist, Brown, Very Dense, Well Graded SAND, Trace Silty Clay, Some Medium to Coarse Sub-angular Gravel (SW-SC)									
		S-3	28-39-32 (71)	Moist, Brown, Very Dense, Well Graded GRAVEL, Few Silt, Some Medium to Coarse Sub-angular Gravel (GW-GM)									
		S-4	50-30-50/2" (50/2")										
		S-5	40-50/5"										
		S-6	12-26-50/4"										
		S-7	50-50/4" (50/4")										
	S-8	18-30-46 (76)											
Bottom of borehole at 16.5-ft.													

CLIENT: AECOM

PROJECT NAME: SEA S Concourse

PROJECT NUMBER: _____

PROJECT LOCATION: _____ SEA _____

DATE STARTED: 05/14/2024 COMPLETED: 05/14/2024

GROUND ELEVATION: 378.54 HOLE SIZE: 10-inch

DRILLING CONTRACTOR: Cascade Drilling

NORTHING: 163865.82 EASTING: 1276875.18

DRILLING METHOD: HSA HAMMER WEIGHT: 140-lbs

GROUND WATER LEVELS: _____

SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch AT TIME OF DRILLING: Not Encountered

LOGGED BY: JA CHECKED BY: AB AT END OF DRILLING: Not Encountered

[illegible]

CLIENT: AECOM

PROJECT NAME: SEA S Concourse

PROJECT NUMBER: _____

PROJECT LOCATION: _____ SEA _____

DATE STARTED: 05/14/2024 COMPLETED: 05/14/2024

GROUND ELEVATION: 378.54 HOLE SIZE: 10-inch

DRILLING CONTRACTOR: Cascade Drilling

NORTHING: 163865.82 EASTING: 1276875.18

DRILLING METHOD: HSA HAMMER WEIGHT: 140-lbs

GROUND WATER LEVELS: _____

SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch AT TIME OF DRILLING: Not Encountered

LOGGED BY: JA CHECKED BY: AB AT END OF DRILLING: Not Encountered

[illegible]

CLIENT:	AECOM		
PROJECT NUMBER:			
DATE STARTED:	05/14/2024	COMPLETED:	05/14/2024
DRILLING CONTRACTOR:	Cascade Drilling		
DRILLING METHOD:	CME 75	HAMMER WEIGHT:	140-lbs
SAMPLING DEVICE:	Split Spoon 2.5-inch	CASING SIZE:	8-inch
LOGGED BY:	AM	CHECKED BY:	AB

PROJECT NAME:	SEA S Concourse		
PROJECT LOCATION:	SEA		
GROUND ELEVATION:	377.94	HOLE SIZE:	12-inch
NORTHING:	163735.73	EASTING:	1276785.11
GROUND WATER LEVELS:			
AT TIME OF DRILLING:	Not Encountered		
AT END OF DRILLING:	Not Encountered		

DEPTH (ft)	SAMPLE TYPE NUMBER	BLOW COUNT (N VALUES)	GRAPHIC LOG	MATERIAL DESCRIPTION	LIQUID LIMIT	PLASTICITY INDEX	% GRAVEL	% SAND	% FINES	RECOVERY (IN)	MOISTURE CONTENT (%)	PID (PPM)	REMARKS
5				14-Inch PCC									
				18-inch Aggregate Base Classified as: Moist, Gray, Fine to Medium Sand, Few Fine Sub-angular Gravel (SP)									
	S-1	10-43-50/4"		Moist, Gray, Very Dense, Silty Medium to Coarse SAND, Little Coarse Sub-angular Gravel (SM)									
	S-2	35-50/4" (50/4")											
	S-3	25-50-50/4"		S-3: Few Coarse Sun-angular Garvel									
	S-4	20-50/4" (50/4")		S-4: Trace Medium to Coarse Sub-angular Gravel									
	S-5	(50/3")											
	S-6	28-50-50/5"											
15	S-7	38-50/4" (50/4")		Moist, Brown, Very Dense, Silty Fine to Medium GRAVEL, Some Fine to Medium Sand (GM)			46.0	42.0	12.0	12	6.0	28.6	
	S-8	(50/5")		S-8: Medium to Coarse Sub-angular Gravel								23.9	
Bottom of borehole at 16.5-ft.													

CLIENT:	AECOM		
PROJECT NUMBER:			
DATE STARTED:	05/14/2024	COMPLETED:	05/14/2024
DRILLING CONTRACTOR:	Cascade Drilling		
DRILLING METHOD:	CME 75	HAMMER WEIGHT:	140-lbs
SAMPLING DEVICE:	Split Spoon 2.5-inch	CASING SIZE:	8-inch
LOGGED BY:	AM	CHECKED BY:	AB

PROJECT NAME:	SEA S Concourse		
PROJECT LOCATION:	SEA		
GROUND ELEVATION:	378.06	HOLE SIZE:	12-inch
NORTHING:	163791.41	EASTING:	1276752.50
GROUND WATER LEVELS:			
AT TIME OF DRILLING:	Not Encountered		
AT END OF DRILLING:	Not Encountered		

DEPTH (ft)	SAMPLE TYPE NUMBER	BLOW COUNT (N VALUES)	GRAPHIC LOG	MATERIAL DESCRIPTION	LIQUID LIMIT	PLASTICITY INDEX	% GRAVEL	% SAND	% FINES	RECOVERY (IN)	MOISTURE CONTENT (%)	PID (PPM)	REMARKS
0 <													

CLIENT:	AECOM		
PROJECT NUMBER:			
DATE STARTED:	05/13/2024	COMPLETED:	05/13/2024
DRILLING CONTRACTOR:	Cascade Drilling		
DRILLING METHOD:	HSA	HAMMER WEIGHT:	140-lbs
SAMPLING DEVICE:	Split Spoon 2.5-inch	CASING SIZE:	8-inch
LOGGED BY:	JA	CHECKED BY:	AB

PROJECT NAME:	SEA S Concourse		
PROJECT LOCATION:	SEA		
GROUND ELEVATION:	377.17	HOLE SIZE:	12-inch
NORTHING:	163612.40	EASTING:	1276641.60
GROUND WATER LEVELS:			
AT TIME OF DRILLING:	NA		
AT END OF DRILLING:	NA		

[illegible]



CLIENT: AECOM
PROJECT NUMBER:
DATE STARTED: 05/13/2024 COMPLETED: 05/13/2024
DRILLING CONTRACTOR: Cascade Drilling
DRILLING METHOD: HSA HAMMER WEIGHT: 140-lbs
SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch
LOGGED BY: JA CHECKED BY: AB

PROJECT NAME: SEA S Concourse
PROJECT LOCATION: SEA
GROUND ELEVATION: 377.57 HOLE SIZE: 12-inch
NORTHING: 163761.90 EASTING: 1276642.40
GROUND WATER LEVELS:
AT TIME OF DRILLING: Not Encountered
AT END OF DRILLING: Not Encountered

DEPTH (ft)	SAMPLE TYPE NUMBER	BLOW COUNT (N VALUES)	GRAPHIC LOG	MATERIAL DESCRIPTION	LIQUID LIMIT	PLASTICITY INDEX	% GRAVEL	% SAND	% FINES	RECOVERY (IN)	MOISTURE CONTENT (%)	PID (PPM)	REMARKS
				14.25-inch Concrete (PCC)									
				18-inch Aggregate Base Classified as: Moist, Gray, Silty Fine to Medium SAND, Little Fine Sub-angular Gravel (SM)			22.9	61.8	15.3		8.8		
	S-1	13-32-25 (57)		Moist, Gray, Very Dense, Silty Fine to Medium SAND, Few Coarse Sub-angular Gravel (SM)						18		2.1	
5	S-2	21-35-41 (76)					16.6	59.6	23.7	18	5.8	2.0	
	S-3	40-50/5" (50/5")		S-3: Fine to Medium Sub-angular Gravel						18		1.4	
	S-4	50/5"								18		1.5	
	S-5	36-45-50/6"								6		1.3	
10	S-6	50-50/4" (50/4")		S-6: Medium to Coarse Sub-angular Gravel						8		0.5	
	S-7	35-50/4" (50/4")								6		0.7	
15	S-8	50/5"		Moist, Gray, Very Dense, Poorly Graded SAND, Few Silt, Some Fine to Medium Gravel (SP-SM)			39.8	48.4	11.7	3	7.0	0.4	

Bottom of borehole at 16.5-ft.

CLIENT: AECOM

PROJECT NAME: SEA S Concourse

PROJECT NUMBER: _____

PROJECT LOCATION: _____ SEA _____

DATE STARTED: 05/13/2024 COMPLETED: 05/13/2024

GROUND ELEVATION: 376.96 HOLE SIZE: 12-inch

DRILLING CONTRACTOR: Cascade Drilling

NORTHING: 163737.22 EASTING: 1276518.05

DRILLING METHOD: HSA HAMMER WEIGHT: 140-lbs

GROUND WATER LEVELS: _____

SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch AT TIME OF DRILLING: Not Encountered

LOGGED BY: EC CHECKED BY: AB AT END OF DRILLING: Not Encountered

[illegible]

CLIENT: AECOM

PROJECT NAME: SEA S Concourse

PROJECT NUMBER: _____

PROJECT LOCATION: _____ SEA _____

DATE STARTED: 05/13/2024 COMPLETED: 05/13/2024

GROUND ELEVATION: 376.96 HOLE SIZE: 12-inch

DRILLING CONTRACTOR: Cascade Drilling

NORTHING: 163737.22 EASTING: 1276518.05

DRILLING METHOD: HSA HAMMER WEIGHT: 140-lbs

GROUND WATER LEVELS: _____

SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch AT TIME OF DRILLING: Not Encountered

LOGGED BY: EC CHECKED BY: AB AT END OF DRILLING: Not Encountered

[illegible]

CLIENT: AECOM

PROJECT NAME: SEA S Concourse

PROJECT NUMBER: _____

PROJECT LOCATION: _____ SEA _____

DATE STARTED: 05/13/2024 COMPLETED: 05/13/2024

GROUND ELEVATION: 376.96 HOLE SIZE: 12-inch

DRILLING CONTRACTOR: Cascade Drilling

NORTHING: 163737.22 EASTING: 1276518.05

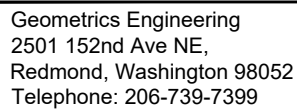
DRILLING METHOD: HSA HAMMER WEIGHT: 140-lbs

GROUND WATER LEVELS: _____

SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch AT TIME OF DRILLING: Not Encountered

LOGGED BY: EC CHECKED BY: AB AT END OF DRILLING: Not Encountered

[illegible]



BORING NUMBER S10-13

PAGE 4 OF 4

CLIENT:	AECOM		
PROJECT NUMBER:			
DATE STARTED:	05/13/2024	COMPLETED:	05/13/2024
DRILLING CONTRACTOR:	Cascade Drilling		
DRILLING METHOD:	HSA	HAMMER WEIGHT:	140-lbs
SAMPLING DEVICE:	Split Spoon 2.5-inch	CASING SIZE:	8-inch
LOGGED BY:	EC	CHECKED BY:	AB

PROJECT NAME:	SEA S Concourse		
PROJECT LOCATION:	SEA		
GROUND ELEVATION:	376.96	HOLE SIZE:	12-inch
NORTHING:	163737.22	EASTING:	1276518.05
GROUND WATER LEVELS:			
AT TIME OF DRILLING:	Not Encountered		
AT END OF DRILLING:	Not Encountered		

[illegible]



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BORING NUMBER S11-15

PAGE 1 OF 1

CLIENT: AECOM
PROJECT NUMBER:
DATE STARTED: 05/15/2024 COMPLETED: 05/15/2024
DRILLING CONTRACTOR: Cascade Drilling
DRILLING METHOD: CME 75 HAMMER WEIGHT: 140-lbs
SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch
LOGGED BY: AM CHECKED BY: AB

PROJECT NAME: SEA S Concourse
PROJECT LOCATION: SEA
GROUND ELEVATION: 376.20 HOLE SIZE: 10-inch
NORTHING: 163643.09 EASTING: 1276369.18
GROUND WATER LEVELS:
AT TIME OF DRILLING: Not Encountered
AT END OF DRILLING: Not Encountered

DEPTH (ft)	SAMPLE TYPE NUMBER	BLOW COUNT (N VALUES)	GRAPHIC LOG	MATERIAL DESCRIPTION	LIQUID LIMIT	PLASTICITY INDEX	% GRAVEL	% SAND	% FINES	RECOVERY (IN)	MOISTURE CONTENT (%)	PID (PPM)	REMARKS
				14.3-inch Concrete (PCC)									
				18-inch Aggregate Base Classified as: Moist, Gray, Fine to Medium Sand, Few Fine Sub-angular Gravel (SP)									
	S-1	14-21-13 (34)		Moist, Gray, Dense to Very Dense, Silty Fine to Medium SAND, Little to Some Fine Sub-Angular Gravel (SM)			21.8	61.6	16.7	18	6.8	0.2	
5	S-2	15-30-30 (60)								18		0	
	S-3	15-25-30 (55)								18		0	
	S-4	14-21-26 (47)					10.0	64.5	25.5	18	8.9	0	
	S-5	18-28-25 (53)								18		1.4	
10	S-6	11-21-22 (43)								18		7.2	
	S-7	28-50/5" (50/5")								10		34	
15	S-8	50/5"					38.0	42.6	19.4	5	5.4	0.5	

Bottom of borehole at 16.5-ft.

CLIENT:	AECOM		
PROJECT NUMBER:			
DATE STARTED:	05/21/2024	COMPLETED:	05/21/2024
DRILLING CONTRACTOR:	Cascade Drilling		
DRILLING METHOD:	CME 75	HAMMER WEIGHT:	140-lbs
SAMPLING DEVICE:	Split Spoon 2.5-inch	CASING SIZE:	8-inch
LOGGED BY:	JA	CHECKED BY:	AB

PROJECT NAME:	SEA S Concourse		
PROJECT LOCATION:	SEA		
GROUND ELEVATION:	375.61	HOLE SIZE:	10-inch
NORTHING:	1635502.07	EASTING:	1276318.05
GROUND WATER LEVELS:			
AT TIME OF DRILLING:	Not Encountered		
AT END OF DRILLING:	Not Encountered		

DEPTH (ft)	SAMPLE TYPE NUMBER	BLOW COUNT (N VALUES)	GRAPHIC LOG	MATERIAL DESCRIPTION	LIQUID LIMIT	PLASTICITY INDEX	% GRAVEL	% SAND	% FINES	RECOVERY (IN)	MOISTURE CONTENT (%)	PID (PPM)	REMARKS
0 <													



CLIENT: AECOM
PROJECT NUMBER:
DATE STARTED: 05/21/2024 COMPLETED: 05/21/2024
DRILLING CONTRACTOR: Cascade Drilling
DRILLING METHOD: HSA HAMMER WEIGHT: 140-lbs
SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch
LOGGED BY: JA CHECKED BY: AB

PROJECT NAME: SEA S Concourse
PROJECT LOCATION: SEA
GROUND ELEVATION: 375.84 HOLE SIZE: 10-inch
NORTHING: 163449.82 EASTING: 1276447.37
GROUND WATER LEVELS:
AT TIME OF DRILLING: Not Encountered
AT END OF DRILLING: Not Encountered

DEPTH (ft)	SAMPLE TYPE NUMBER	BLOW COUNT (N VALUES)	GRAPHIC LOG	MATERIAL DESCRIPTION	LIQUID LIMIT	PLASTICITY INDEX	% GRAVEL	% SAND	% FINES	RECOVERY (IN)	MOISTURE CONTENT (%)	PID (PPM)	REMARKS
				19.5-Inch PCC									
	S-1	20-28-17 (45)		18-inch Aggregate Base Classified as: Moist, Gray, Fine to Medium Sand, Few Fine Sub-angular Gravel (SP)						18		3.7	
5	S-2	6-6-11 (17)		Moist, Brown-Gray, Medium Dense, Silty Fine to Medium SAND, Little Fine Sub-angular Gravel (SM)			21.2	53.4	25.4	18	11.8	2.1	Slight Glycol Odor
	S-3	10-14-16 (30)							19.4	12	7.9	2.6	
	S-4	17-20-27 (47)		S-4: Gray, Dense						18		1.8	
10	S-5	20-30-36 (66)		S-4: Very Dense to 16.5-ft, Few Sub-angular Cobble			18.6	53.6	23.8	12	8	1.6	
	S-6	24-23-50/5" (50/5")								16		1.9	
	S-7	50-50/4" (50/4")		S-7: Brown						10		0.7	
15	S-8	22-40-42 (82)		S-8: Coarse Sub-rounded Gravel			16.9	53.0	30.1	18	8.4	0.0	

Bottom of borehole at 16.5-ft.

CLIENT:	AECOM		
PROJECT NUMBER:			
DATE STARTED:	05/21/2024	COMPLETED:	05/22/2024
DRILLING CONTRACTOR:	ConeTech		
DRILLING METHOD:	Mud Rotary	HAMMER WEIGHT:	140-lbs
SAMPLING DEVICE:	Split Spoon 2.5-inch	CASING SIZE:	8-inch
LOGGED BY:	AM	CHECKED BY:	AB

PROJECT NAME:	SEA S Concourse		
PROJECT LOCATION:	SEA		
GROUND ELEVATION:	376.12	HOLE SIZE:	10-inch
NORTHING:	163437.87	EASTING:	1276533.08
GROUND WATER LEVELS:			
AT TIME OF DRILLING:	Not Encountered		
AT END OF DRILLING:	Not Encountered		

[illegible]

CLIENT: AECOM

PROJECT NAME: SEA S Concourse

PROJECT NUMBER: _____

PROJECT LOCATION: _____ SEA _____

DATE STARTED: 05/21/2024 COMPLETED: 05/22/2024

GROUND ELEVATION: 376.12 HOLE SIZE: 10-inch

DRILLING CONTRACTOR: Cascade Drilling

NORTHING: 163437.87 EASTING: 1276533.08

DRILLING METHOD: HSA HAMMER WEIGHT: 140-lbs

GROUND WATER LEVELS: _____

SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch AT TIME OF DRILLING: Not Encountered

LOGGED BY: JA CHECKED BY: AB AT END OF DRILLING: Not Encountered

[illegible]

CLIENT: AECOM

PROJECT NAME: SEA S Concourse

PROJECT NUMBER: _____

PROJECT LOCATION: _____ SEA

DATE STARTED: 05/21/2024 COMPLETED: 05/22/2024

GROUND ELEVATION: 376.12 HOLE SIZE: 10-inch

DRILLING CONTRACTOR: Cascade Drilling

NORTHING: 163437.87 EASTING: 1276533.08

DRILLING METHOD: HSA HAMMER WEIGHT: 140-lbs

GROUND WATER LEVELS: _____

SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch AT TIME OF DRILLING: Not Encountered

LOGGED BY: JA CHECKED BY: AB AT END OF DRILLING: Not Encountered

[illegible]

CLIENT: AECOM

PROJECT NAME: SEA S Concourse

PROJECT NUMBER: _____

PROJECT LOCATION: _____ SEA _____

DATE STARTED: 05/21/2024 COMPLETED: 05/22/2024

GROUND ELEVATION: 376.12 HOLE SIZE: 10-inch

DRILLING CONTRACTOR: Cascade Drilling

NORTHING: 163437.87 EASTING: 1276533.08

DRILLING METHOD: HSA HAMMER WEIGHT: 140-lbs

GROUND WATER LEVELS: _____

SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch AT TIME OF DRILLING: Not Encountered

LOGGED BY: JA CHECKED BY: AB AT END OF DRILLING: Not Encountered

[illegible]



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BORING NUMBER S12-19

PAGE 1 OF 1

CLIENT: AECOM
PROJECT NUMBER:
DATE STARTED: 05/21/2024 COMPLETED: 05/21/2024
DRILLING CONTRACTOR: Cascade Drilling
DRILLING METHOD: CME 75 HAMMER WEIGHT: 140-lbs
SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch
LOGGED BY: JA CHECKED BY: AB

PROJECT NAME: SEA S Concourse
PROJECT LOCATION: SEA
GROUND ELEVATION: 375.37 HOLE SIZE: 10-inch
NORTHING: 163348.67 EASTING: 1276387.68
GROUND WATER LEVELS:
AT TIME OF DRILLING: Not Encountered
AT END OF DRILLING: Not Encountered

DEPTH (ft)	SAMPLE TYPE NUMBER	BLOW COUNT (N VALUES)	GRAPHIC LOG	MATERIAL DESCRIPTION	LIQUID LIMIT	PLASTICITY INDEX	% GRAVEL	% SAND	% FINES	RECOVERY (IN)	MOISTURE CONTENT (%)	PID (PPM)	REMARKS
				14.75-Inch PCC									
				18-inch Aggregate Base Classified as: Moist, Gray, Fine to Medium Sand, Few Fine Sub-angular Gravel (SP)									
	S-1	13-37-50/4.5" (50/4.5")		Moist, Gray, Very Dense, Poorly Graded GRAVEL, Few Fine to Medium Silt, Mostly Fine to Medium Sand (GP-GM)						12		24	Slight Glycol Odor
5	S-2	23-50/5.5" (50/5.5")		S-2: Few Silt			45.3	42.2	10.2	12	7.8	34	Slight Glycol Odor
	S-3	17-30-20 (50)		Moist, Gray, Dense, Silty Medium to Coarse Sand, Few Fine to Medium Sub-rounded Gravel (SM)						18		140	
	S-4	10-16-24 (40)								18		6.8	
	S-5	20-50/5" (50/5")								3		0.8	
10	S-6	17-31-47 (78)							34.8	16	10	22	Slight Glycol Odor
	S-7	40-50/5" (50/5")		S-7: Brown						8		20	
15	S-8	(50/5")		S8: Some Fine to Medium Sub-rounded Gravel			32.2	43.2	24.6	4	5.5	9.3	

Bottom of borehole at 16.5-ft.



CLIENT: AECOM
PROJECT NUMBER:
DATE STARTED: 05/22/2024 COMPLETED: 05/22/2024
DRILLING CONTRACTOR: Cascade Drilling
DRILLING METHOD: CME 75 HAMMER WEIGHT: 140-lbs
SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch
LOGGED BY: AM CHECKED BY: AB

PROJECT NAME: SEA S Concourse
PROJECT LOCATION: SEA
GROUND ELEVATION: 374.82 HOLE SIZE: 10-inch
NORTHING: 163289.55 EASTING: 1276285.04
GROUND WATER LEVELS:
AT TIME OF DRILLING: Not Encountered
AT END OF DRILLING: Not Encountered

DEPTH (ft)	SAMPLE TYPE NUMBER	BLOW COUNT (N VALUES)	GRAPHIC LOG	MATERIAL DESCRIPTION	LIQUID LIMIT	PLASTICITY INDEX	% GRAVEL	% SAND	% FINES	RECOVERY (IN)	MOISTURE CONTENT (%)	PID (PPM)	REMARKS
				17.3-Inch PCC									
	S-1	21-24-25 (49)		18-inch Aggregate Base Classified as: Moist, Gray, Fine to Medium Sand, Few Fine Sub-angular Gravel (SP)			32.7	60.7	6.6	18	6.5	0.0	
5	S-2	24-24-24 (48)		Moist, Gray, Dense, Poorly Graded SAND, Trace Fine to Medium Silt, Little Fine to Medium Sub-Angular Gravel (SP-SM)						14		0.0	
	S-3	10-9-13 (22)		Moist, Gray, Medium Dense, Well Graded SAND, Few Fine to Medium Silt, Some Fine to Medium Sub-Angular Gravel (SW-SM)			39.6	49	11.4	14	6.5	0.0	
	S-4	18-19-17 (36)		Moist, Gray, Dense to Very Dense, Silty Fine to Medium Sand, Little Fine to Medium Sub-angular Gravel (SM)						18		0.0	
	S-5	12-14-14 (28)		S-5: Medium Dense to 12-ft					27.3	18	8.9	0.0	
10	S-6	12-10-9 (19)		S-6: Some Fine to Medium Sub-angular Gravel						18		0.0	
	S-7	40-50/5" (50/5")								14		0.0	
15	S-8	(50/5")								2		0.0	

Bottom of borehole at 16.5-ft.



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BORING NUMBER S15-21

PAGE 1 OF 1

CLIENT: AECOM
PROJECT NUMBER:
DATE STARTED: 05/22/2024 COMPLETED: 05/22/2024
DRILLING CONTRACTOR: Cascade Drilling
DRILLING METHOD: CME 75 HAMMER WEIGHT: 140-lbs
SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch
LOGGED BY: AM CHECKED BY: AB

PROJECT NAME: SEA S Concourse
PROJECT LOCATION: SEA
GROUND ELEVATION: 375.55 HOLE SIZE: 10-inch
NORTHING: 163312.73 EASTING: 1276518.51
GROUND WATER LEVELS:
AT TIME OF DRILLING: Not Encountered
AT END OF DRILLING: Not Encountered

DEPTH (ft)	SAMPLE TYPE NUMBER	BLOW COUNT (N VALUES)	GRAPHIC LOG	MATERIAL DESCRIPTION	LIQUID LIMIT	PLASTICITY INDEX	% GRAVEL	% SAND	% FINES	RECOVERY (IN)	MOISTURE CONTENT (%)	PID (PPM)	REMARKS
				14-Inch PCC									
				18-inch Aggregate Base Classified as:									
				Moist, Gray, Fine to Medium Sand, Few Fine Sub-angular Gravel (SP)									
	S-1	26-32-33 (65)		Moist, Brown, Very Dense, Fine to Medium Sand, Some Fine to Medium Sub-angular Gravel (SM)						18		0.0	
5	S-2	18-17-22 (39)		Moist, Gray, Dense, Well Graded GRAVEL, Few Fine to Medium Silt, Some Fine to Medium Sand (GW-GM)			47	43.7	9.3	12	4.22	0.0	
	S-3	12-16-23 (39)		Moist, Brown, Dense, Silty Fine to Medium Sand, Few Fine to Medium Sub-angular Gravel (SM)						18		0.0	
	S-4	4-8-18 (26)								18		0.0	
	S-5	14-21-29 (50)					12.3	62.1	25.6	18	7.1	0.0	
10	S-6	14-27-28 (55)								18		0.0	
	S-7	20-30-42 (72)							29.5	18	8.7	0.0	
15	S-8	14-33-42 (75)								18		0.0	

Bottom of borehole at 16.5-ft.



CLIENT: AECOM
PROJECT NUMBER:
DATE STARTED: 05/22/2024 COMPLETED: 05/22/2024
DRILLING CONTRACTOR: Cascade Drilling
DRILLING METHOD: CME 75 HAMMER WEIGHT: 140-lbs
SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch
LOGGED BY: AM CHECKED BY: AB

PROJECT NAME: SEA S Concourse
PROJECT LOCATION: SEA
GROUND ELEVATION: 374.73 HOLE SIZE: 10-inch
NORTHING: 163150.71 EASTING: 1276389.65
GROUND WATER LEVELS:
AT TIME OF DRILLING: 8-Feet
AT END OF DRILLING: 8-Feet

DEPTH (ft)	SAMPLE TYPE NUMBER	BLOW COUNT (N VALUES)	GRAPHIC LOG	MATERIAL DESCRIPTION	LIQUID LIMIT	PLASTICITY INDEX	% GRAVEL	% SAND	% FINES	RECOVERY (IN)	MOISTURE CONTENT (%)	PID (PPM)	REMARKS
				14.5-Inch PCC									
				18-inch Aggregate Base Classified as: Moist, Gray, Fine to Medium Sand, Few Fine Sub-angular Gravel (SP)									
	S-1	22-24-22 (46)		Moist, Gray, Dense, Silty Fine to Medium SAND, Some Fine to Medium Sub-angular Gravel (SM)			26.4	58.1	15.6	18	6.7	0.6	Slightly Glycol Odour
5	S-2	19-23-20 (43)		S-2: Some Fine to Medium Angular to Sub-angular Gravel						10		0.5	Slightly Glycol Odour
	S-3	6-13-11 (24)		S-3: Medium Dense, Some Fine to Medium Sub-angular to Sub-rounded Gravel						10		0.7	Slightly Glycol Odour
	S-4	14-18-12 (30)		Moist, Brown, Dense to Very Dense, Poorly Graded SAND, Few Fine to Medium Silt, Some Fine to Medium Sub-Angular Gravel (SP-SM)			43.3	45.4	11.3	14	11.6	0.2	
10	S-5	9-13-20 (33)								16		2.6	
	S-6	19-28-50/5" (50/5")								18		9.8	
	S-7	42-50-50/3"		Moist, Gray, Very Dense, Silty Fine to Medium SAND, Trace Fine to Medium Sub-Angular Gravel (SM)			9	59.8	31.1	18	9.3	98.0	
15													
	S-8	50/5" (50/5")								6		51.5	

Bottom of borehole at 16.5-ft.



CLIENT: AECOM
PROJECT NUMBER:
DATE STARTED: 05/20/2024 COMPLETED: 05/20/2024
DRILLING CONTRACTOR: Cascade Drilling
DRILLING METHOD: CME 75 HAMMER WEIGHT: 140-lbs
SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch
LOGGED BY: JA CHECKED BY: AB

PROJECT NAME: SEA S Concourse
PROJECT LOCATION: SEA
GROUND ELEVATION: 374.61 HOLE SIZE: 10-inch
NORTHING: 163114.14 EASTING: 1276592.40
GROUND WATER LEVELS:
AT TIME OF DRILLING: Not Encountered
AT END OF DRILLING: Not Encountered

DEPTH (ft)	SAMPLE TYPE NUMBER	BLOW COUNT (N VALUES)	GRAPHIC LOG	MATERIAL DESCRIPTION	LIQUID LIMIT	PLASTICITY INDEX	% GRAVEL	% SAND	% FINES	RECOVERY (IN)	MOISTURE CONTENT (%)	PID (PPM)	REMARKS
				15.5-Inch PCC									
				18-Inch Aggregate Base Classified As: Moist, Gray, Fine Sub-rounded Gravel with Fine to Medium Sand (GP)									
	S-1	23-24-20 (44)		Moist, Gray, Dense to Very Dense, Silty Fine to Medium SAND, Few Fine Sub-rounded Gravel (SM)			11.4	62.8	25.8	18	7.0	0.4	
5	S-2	15-15-14 (29)		S-2: Medium Dense to 12-ft						18		1.8	
	S-3	10-11-9 (19)								18		1.8	
	S-4	8-13-14 (27)					11.8	56.8	31.4	18	11.5	1.5	
	S-5	6-15-14 (29)								18		1.0	
10	S-6	8-12-11 (23)								12		00	
	S-7	8-16-25 (41)								16		00	
15	S-8	12-32-50/3" (50/3")		S-8: Few Sub-angular Cobble			16.6	55.2	28.2	12	8.0	2.8	

Bottom of borehole at 16.5-ft.

CLIENT:	AECOM		
PROJECT NUMBER:			
DATE STARTED:	05/20/2024	COMPLETED:	05/21/2024
DRILLING CONTRACTOR:	Cascade Drilling		
DRILLING METHOD:	HSA	HAMMER WEIGHT:	140-lbs
SAMPLING DEVICE:	Split Spoon 2.5-inch	CASING SIZE:	8-inch
LOGGED BY:	JA	CHECKED BY:	AB

PROJECT NAME:	SEA S Concourse		
PROJECT LOCATION:	SEA		
GROUND ELEVATION:	375.86	HOLE SIZE:	10-inch
NORTHING:	163262.08	EASTING:	1276667.26
GROUND WATER LEVELS:			
AT TIME OF DRILLING:	Not Encountered		
AT END OF DRILLING:	Not Encountered		

[illegible]

CLIENT: AECOM

PROJECT NAME: SEA S Concourse

PROJECT NUMBER: _____

PROJECT LOCATION: _____ SEA _____

DATE STARTED: 05/20/2024 COMPLETED: 05/21/2024

GROUND ELEVATION: 375.86 HOLE SIZE: 10-inch

DRILLING CONTRACTOR: Cascade Drilling

NORTHING: 163262.08 EASTING: 1276667.26

DRILLING METHOD: HSA HAMMER WEIGHT: 140-lbs

GROUND WATER LEVELS: _____

SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch AT TIME OF DRILLING: Not Encountered

LOGGED BY: JA CHECKED BY: AB AT END OF DRILLING: Not Encountered

[illegible]



CLIENT: AECOM
PROJECT NUMBER:
DATE STARTED: 05/20/2024 COMPLETED: 05/20/2024
DRILLING CONTRACTOR: Cascade Drilling
DRILLING METHOD: CME 75 HAMMER WEIGHT: 140-lbs
SAMPLING DEVICE: Split Spoon 2.5-inch CASING SIZE: 8-inch
LOGGED BY: JA CHECKED BY: AB

PROJECT NAME: SEA S Concourse
PROJECT LOCATION: SEA
GROUND ELEVATION: 375.13 HOLE SIZE: 10-inch
NORTHING: 163145.34 EASTING: 1276734.10
GROUND WATER LEVELS:
AT TIME OF DRILLING: Not Encountered
AT END OF DRILLING: Not Encountered

DEPTH (ft)	SAMPLE TYPE NUMBER	BLOW COUNT (N VALUES)	GRAPHIC LOG	MATERIAL DESCRIPTION	LIQUID LIMIT	PLASTICITY INDEX	% GRAVEL	% SAND	% FINES	RECOVERY (IN)	MOISTURE CONTENT (%)	PID (PPM)	REMARKS
				14.6-Inch PCC									
				18-inch Aggregate Base Classified as: Moist, Gray, Medium to Coarse Sand, Little Fine to Medium Sub-angular Gravel (SP)									
	S-1	13-12-9 (21)		Moist, Gray, Medium Dense, Silty Fine to Medium Sand, Few Fine Sub-angular Gravel (SM)						12		0.4	
5	S-2	13-12-19 (31)		S-2: Little Fine Sub-angular Gravel			14.1	62.4	23.5	18	5.4	0.6	
	S-3	16-27-26 (53)		S-3: Few Fine Sub-angular Gravel						16		0.0	
	S-4	23-24-25 (49)		S-4: Brown, Dense, Few Coarse Sub-angular Gravel			31.0	52.3	16.7	16	7.4	0.0	
	S-5	16-28-25 (53)		S-5: Few Fine Sub-angular Gravel						18		0.0	
10	S-6	23-24-24 (48)								18		0.0	
	S-7	22-33-50 (83)								19		0.1	
15	S-8	23-46-34 (80)		S-8: Little Coarse Sub-angular Gravel			31.0	52.3	16.7	18	6.6	0.0	

Bottom of borehole at 16.5-ft.